DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

GROUP : 56

PROJECT STATEMENT: **BRAIN TUMOUR DATA AUGMENTATION USING GENERATIVE ADVERSARIAL NETWORKS (GAN) AND SEGMENTATION USING U-NET ARCHITECTURE.**

Project II Presentation PROJCS781

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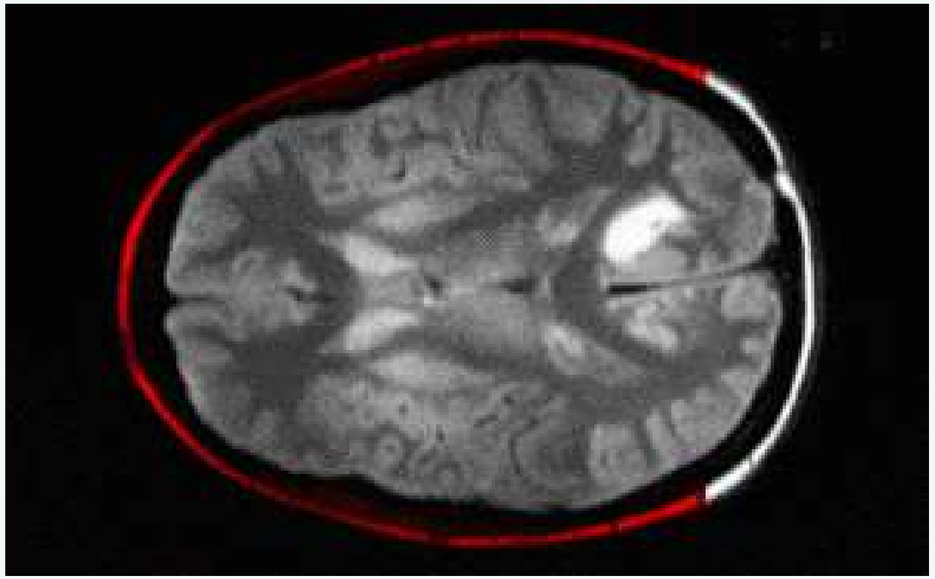
Presented By: Tirthankar Sardar
Sriparna Debbarman
Shamik Sutradhar
Debaditya Bhattacharjee

Under the guidance of - Prof. Antara Das

BRIEF INTRODUCTION:

- Medical image analysis is critical for diagnosis, treatment planning, and biomedical research
- A key task is image segmentation — assigning a label to every pixel
- Traditional methods struggle with accuracy and limited annotated data
- Deep learning, especially Convolutional Neural Networks (CNNs), has transformed image analysis
- U-Net is a specialized CNN architecture designed for biomedical image segmentation
- It achieves precise localization even with very few training images

PROJECT DEFINITION:



THIS PROJECT INVOLVES AUTOMATED BRAIN TUMOUR SEGMENTATION USING DEEP LEARNING AND COMPUTER VISION TO ACCURATELY DELINEATE TUMOR REGIONS FROM MULTIMODAL MRI SCANS. THE CORE METHODOLOGY EMPLOYS THE U-NET CONVOLUTIONAL NEURAL NETWORK, WHICH UTILIZES AN ENCODER-DECODER STRUCTURE WITH SKIP CONNECTIONS TO ACHIEVE HIGH-LEVEL CONTEXTUAL UNDERSTANDING AND PRECISE SPATIAL LOCALIZATION.

THIS PROCESS, WHICH INVOLVES KEY STEPS LIKE IMAGE PREPROCESSING AND SEMANTIC SEGMENTATION, IS CRUCIAL FOR IMPROVING DIAGNOSIS AND TREATMENT PLANNING WHILE REDUCING THE MANUAL WORKLOAD FOR MEDICAL PROFESSIONALS.

Technical Concepts

- Convolutional Neural Networks (CNNs)
- Semantic Segmentation
- Pixel-level Classification
- Image Preprocessing (normalization, noise reduction)
- Data Augmentation

IMPORTANCE OF GENERATIVE ADVERSARIAL NETWORKS

- In India, around 40,000–50,000 brain tumor cases are diagnosed annually.
- This corresponds to approximately 0.0035% of the total population (1.417 billion).

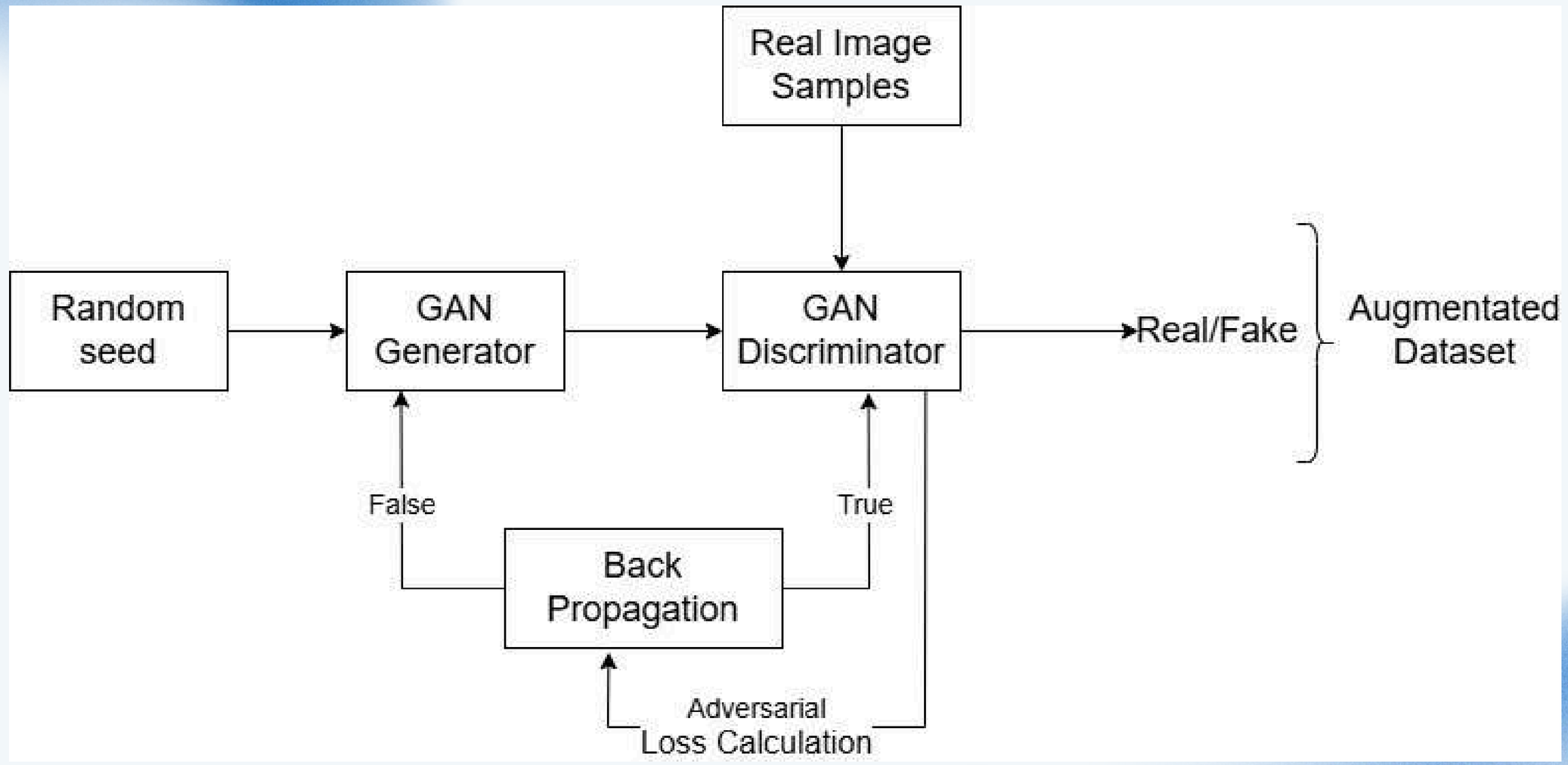
- Assuming MRI scans are 100% accurate:
  - For every 10,000 MRI scans, only about 35 show brain tumors.
  - The remaining samples belong to the non-tumor (majority) class.
- This results in severe class imbalance in medical imaging datasets.

GENERATIVE(DEEP GENERATIVE) MODELS LEARN THE UNDERLYING DATA DISTRIBUTION.
THEY CAN SYNTHESIZE REALISTIC AND
DIVERSE TUMOR IMAGES.

MODELS BECOME BIASED TOWARD THE MAJORITY (NON-TUMOR) CLASS.REDUCED SENSITIVITY AND HIGHER FALSE-NEGATIVE RATES FOR TUMOR DETECTION, COMBINED WITH LIMITED AVAILABILITY AND RESTRICTED ACCESS TO LABELED MEDICAL DATA, FURTHER AMPLIFY BIAS.

Synthetic data augmentation helps balance classes.
Improves model robustness and generalization.
Enhances performance of downstream diagnostic and classification models.

BLOCK DIAGRAM OF GENERATIVE ADVERSARIAL NETWORKS



IMPORTANCE AND ADVANTAGES OF U-NET ARCHITECTURE

CLINICAL IMPORTANCE

Accurate localization of tumor regions for diagnosis and treatment planning.
Helps in monitoring tumor growth and treatment response.

U-NET ARCHITECTURE

- Encoder-decoder CNN with skip connections.
- Contracting path consists of Convolution and pooling layers for capturing features and down sampling. Expanding path consists of up-convolution for generating segmentation map.
- Combines high-level features with fine spatial details.
- Efficient for pixel-wise prediction.

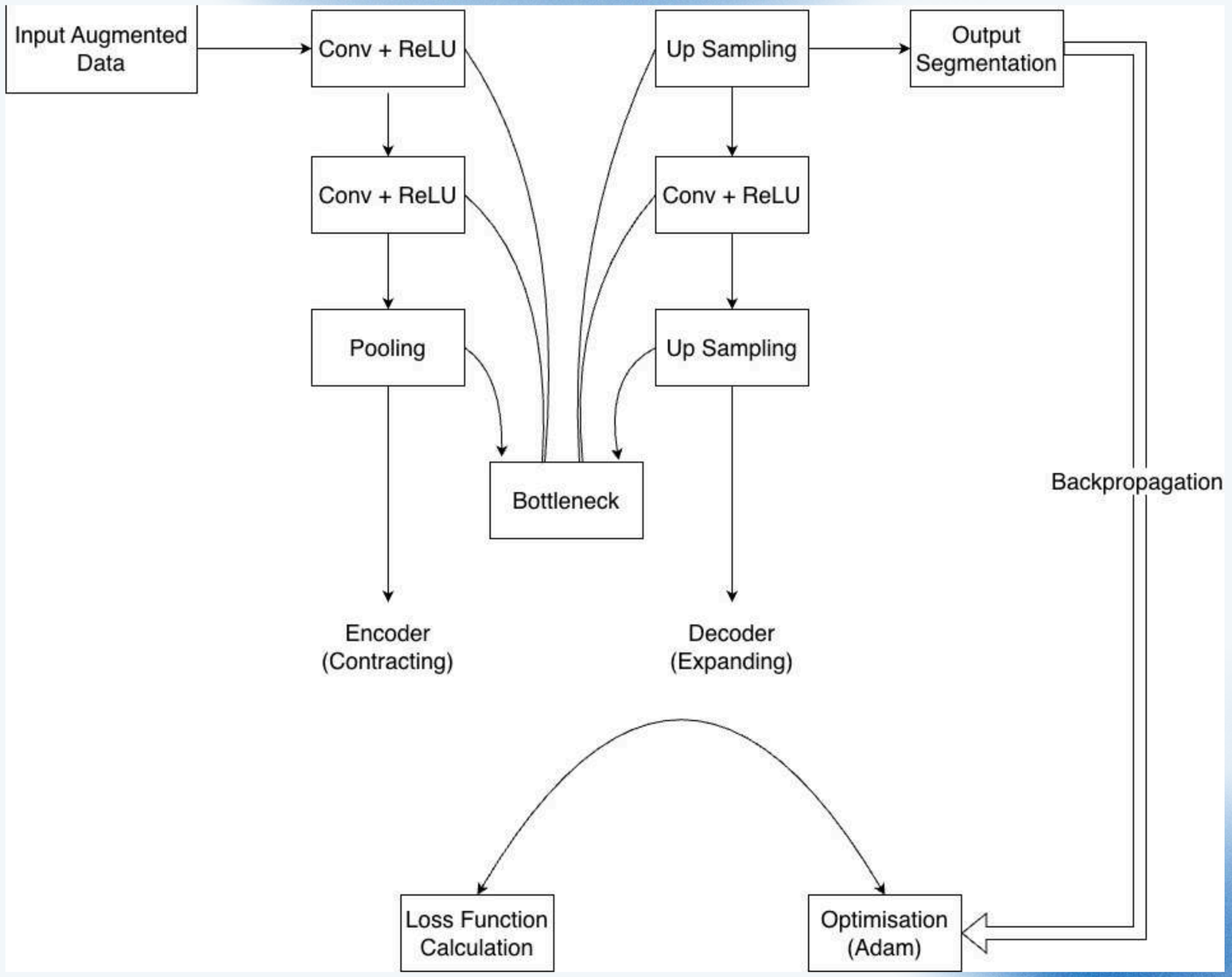
ADVANTAGES

- WORKS WELL WITH LIMITED MEDICAL IMAGING DATA.
- PRESERVES SPATIAL INFORMATION FOR PRECISE SEGMENTATION.
- LEARNS MULTI-SCALE FEATURES EFFECTIVELY.
- SUPPORTS END-TO-END TRAINING.
- HANDLES CLASS IMBALANCE USING DICE OR WEIGHTED LOSS.
- COMPUTATIONALLY EFFICIENT AND FAST.

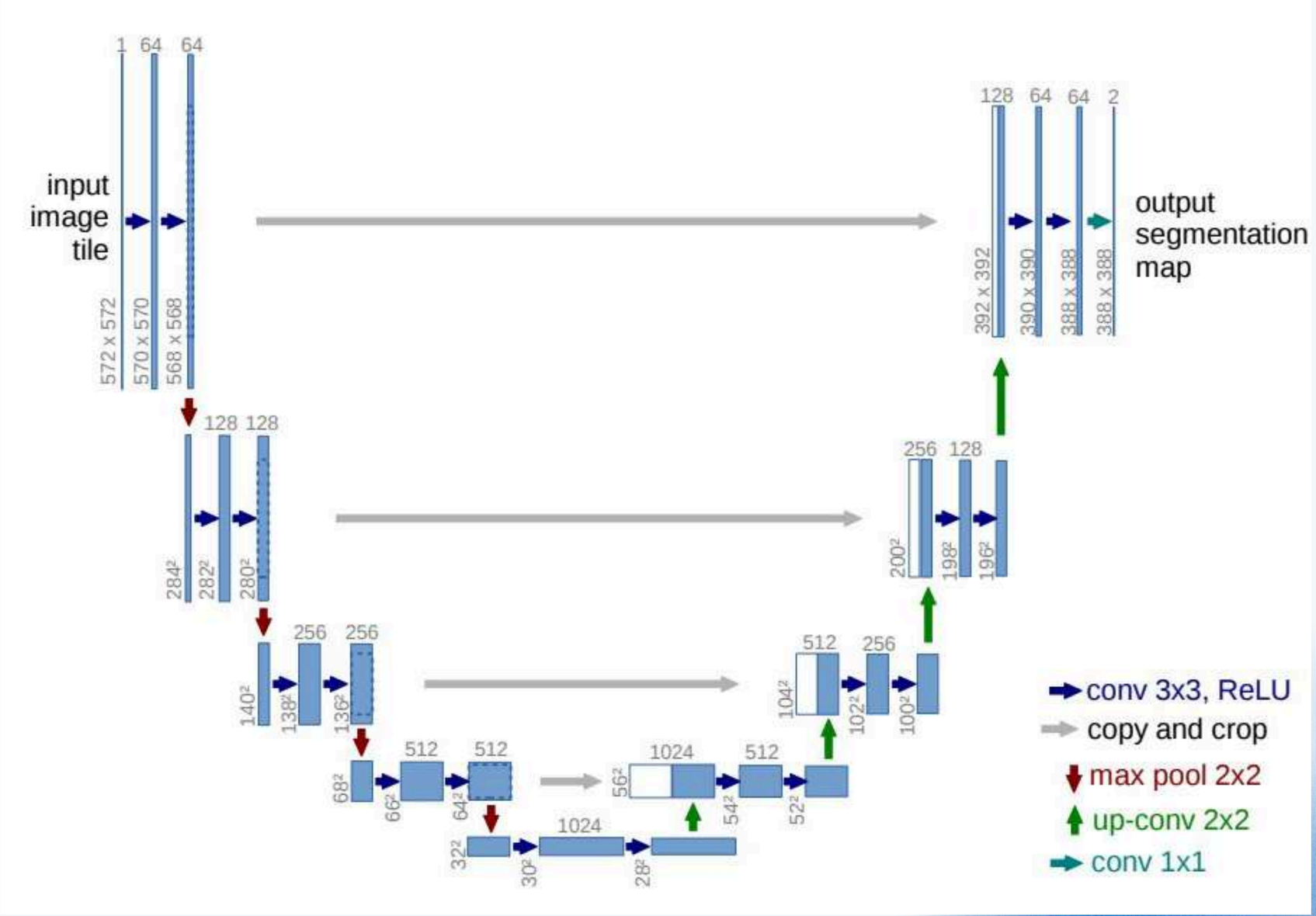
WHY SEGMENTATION ?

- PROVIDES PIXEL-LEVEL TUMOR BOUNDARIES INSTEAD OF JUST CLASS LABELS.
- ENABLES MEASUREMENT OF TUMOR SIZE, SHAPE, AND VOLUME.

BLOCK DIAGRAM OF U-NET ARCHITECTURE

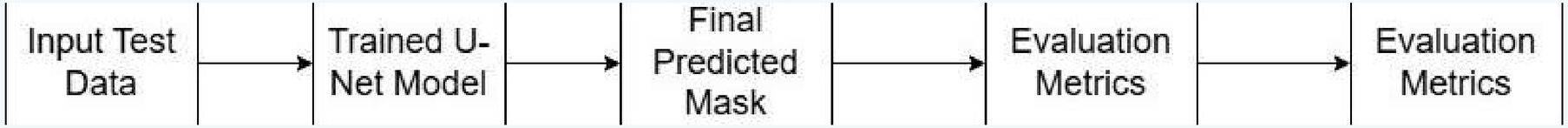


U-NET ARCHITECTURE

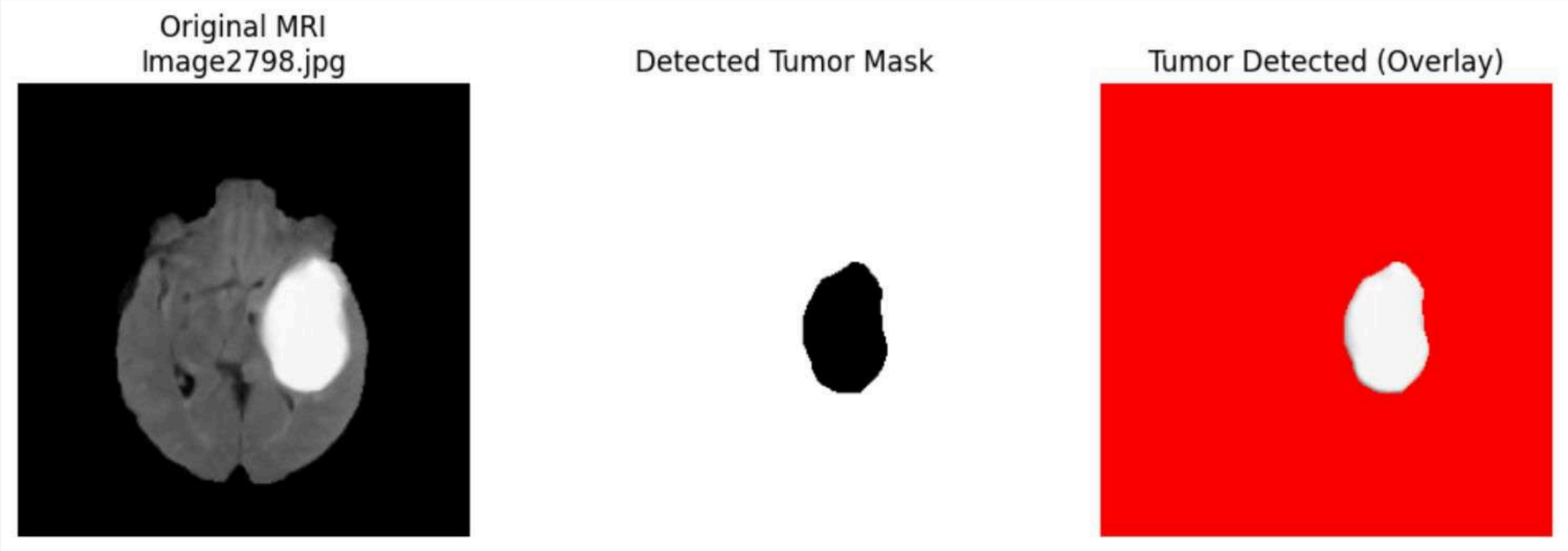


EVALUATION STAGES

The Testing and Validation stage serves as the final audit of my project. After the GAN has expanded our dataset and the U-Net has completed its iterative training process, we must "lock" the model's weights and test it on a separate hold-out dataset. This ensures that the high accuracy achieved during training wasn't just due to the model memorizing the GAN-generated images, but rather because it truly understands the biological features of a brain tumor.



SEGMENTATION OUTPUT :



Future Scope

- More advanced data generation techniques such as Conditional GANs or diffusion models can be used to create higher-quality synthetic tumor images.

- The basic U-Net model can be improved using advanced versions like Attention U-Net and Residual U-Net to achieve more accurate tumor segmentation.

- Incorporate Vision Transformers (ViT) or hybrid CNN and Transformer models, like U-Net Transformer. This improves global feature representation and long-range dependency modeling in MRI scans.

- Move from 2D slice-based segmentation to 3D U-Net architectures for volumetric MRI analysis. This allows for better spatial consistency across slices.

CONCLUSION

This project presents an effective brain tumor segmentation approach using the U-Net architecture combined with GAN-based data augmentation. U-Net enables accurate tumor region detection from MRI images, while GAN-generated synthetic data helps address data imbalance and improves model robustness.

The results demonstrate that this integrated deep learning framework enhances segmentation performance and has strong potential for supporting automated and reliable medical image analysis in clinical applications.

THANK YOU

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